

Pratkar — Product Requirements Document

Health insurance claim rejection contest engine · Geek2Code 2026

2026-09-17 ·

1. Document Overview

This PRD specifies the MVP that Pratkar will demonstrate at the Geek2Code Grand Final on 20 September 2026. It converts the finalized submission deck into build-ready requirements without changing the problem, users, solution, business model or scope established there.

Field	Value
Project name	Pratkar (health insurance claim rejection contest engine)
Version	1.0
Date	17 September 2026
Status	MVP in development. Mentoring round today; Grand Final 20 September 2026
Theme	FinTech / InsurTech, Open Innovation track
Team	Sid, Pratham, Navya, Parnika, Dhiraj
Source of truth	Finalized Geek2Code submission deck (Pratkar_Geek2Code.pptx)

Purpose. Give the build team one agreed definition of what ships, so that scope arguments are settled in this document rather than during the last 72 hours of development.

Intended audience. The five-member build team; hackathon mentors and judges reviewing technical depth; any contributor joining after the hackathon.

Convention used throughout. Every requirement is tagged by source: **[Deck]** = stated in the finalized deck · **[Rec]** = our technical recommendation, not yet in the deck · **[TBD]** = requires a team decision before build.

2. Product Overview

Pratkar tells an Indian policyholder whether a rejected health insurance claim can be contested, and drafts the appeal.

The user uploads two documents they already have — the insurer's rejection letter and their own policy wording PDF. The system extracts the stated rejection ground and the clause the insurer cited, retrieves that clause verbatim from the policy, tests it against the policy's own wording and against published IRDAI provisions, and returns a contestability verdict of Strong, Moderate or Weak. Every statement in the output links to its source. Where the verdict supports it, the system generates a ready-to-file appeal in English or Hindi.

Problem. A rejection letter cites a clause. Verifying whether that clause actually applies means reading a 40–80 page policy wording against the letter and against IRDAI regulation — skilled legal work almost no policyholder can perform. **[Deck]**

Why it matters. In FY25, 87.50% of health claims were settled by number, meaning roughly one in eight went unpaid (IRDAI, via Ministry of Finance, March 2026). Health insurance accounted for 61.5% of all Insurance Ombudsman complaints — 32,655 of 53,102 (Council for Insurance Ombudsmen, Annual Report 2024-25). Of the complaints decided by formal award, 9,234 of 18,171 — 50.8% — went in the policyholder's favour. Almost nobody contests; roughly half of those who do, win. Households still pay 43.4% of India's total health spending out of pocket (National Health Accounts 2022-23, released May 2026). **[Deck]**

Product vision. Make contesting an unjust claim rejection something a policyholder can do in an evening, without a lawyer and without a fee — starting with health insurance, extending later to motor and life on the same architecture.

3. Problem Definition

The barrier to contesting a claim rejection is comprehension, not merit.

Current problem. A policyholder receives a rejection letter citing a clause — for example "Clause 4.2, pre-existing disease." They cannot tell whether that clause applies to their circumstances, so they accept the decision and absorb the loss.

Root causes

1. **Asymmetry of expertise.** The insurer employs people who read policy wordings professionally. The policyholder does not.
2. **Document length and language.** A health policy wording runs 40–80 pages of cross-referenced definitions, exclusions, waiting periods and sub-limits.

3. **A second rulebook the policyholder has never seen.** Some grounds are invalid under IRDAI regulation regardless of what the policy says — the 60-month moratorium being the clearest example. Policyholders do not know these provisions exist.
4. **No feedback loop.** Insurance Ombudsman awards are released only to the individual complainant, so no public body of precedent accumulates that a policyholder could consult.

Existing workflow

flowchart LR

A[Claim filed] --> B[Rejection letter
cites a clause]

B --> C{Can the family
assess it?}

C -->|Almost always no| D[Loss absorbed]

C -->|Rarely| E[Pay a consultant
or self-file]

E --> F[Grievance officer]

F --> G[Ombudsman]

Pain points

Who	Pain
Policyholder	Cannot judge whether the cited clause applies; no idea a regulatory rulebook exists; unaware of the one-year Ombudsman limitation
Policyholder	Paid consultancy is the only assisted route — a cost borne at the worst possible moment
Insurance agent	Fields these questions from clients with no tool to answer them
Hospital TPA desk	Handles rejections in volume, manually, case by case

Consequences. The household absorbs a cost it may not owe. Grounds that would not survive examination are never examined, so they keep being used. Out-of-pocket health spending stays high.

Why existing approaches are insufficient. The leading claim-advocacy service, Insurance Samadhan, has handled 20,000+ grievances since 2018 at ₹999 plus a 20% success fee **[Deck]**. That is skilled human labour, and expert hours are the binding constraint — the model cannot scale to the volume of rejections. Generic AI document tools have no regulatory rulebook and no obligation to ground their assertions in the user's own policy.

4. Target Users

The primary user is an Indian policyholder whose health insurance claim has been rejected or partially settled.

Role	Type	Objective	Pain point
Policyholder or family member	Primary	Find out whether the rejection can be challenged, and act before the deadline	Cannot read a policy wording against a rejection letter; does not know IRDAI provisions exist
Insurance agent or broker	Secondary [Deck]	Answer a client's rejection question quickly and credibly	No tool; relies on personal experience
Hospital TPA or billing desk	Secondary [Deck]	Resolve rejections in volume for admitted patients	Manual, case by case, no regulatory check
Employer group-policy administrator	Secondary [Deck]	Support employees whose group claims are denied	Same problem at organisational scale
Legal-aid or consumer NGO	Secondary [Deck]	Serve more complainants with the same staff	Expert hours are the constraint

Stakeholders not using the product. Insurers and TPAs are the counterparty to the user, not customers — this is deliberate and is the basis of the competitive position in §6 of the BRD. IRDAI and the Insurance Ombudsman are the institutions whose published rules and process the product operates within; neither is integrated with and neither has been approached. **[Deck]**

MVP user assumption requiring validation. The MVP assumes the user can produce both the rejection letter and the policy wording PDF. Where an insurer's rejection is a terse SMS or email citing no clause, the primary flow degrades — handling is specified in FR-11. **[Rec]**

5. Product Goals

The hackathon goal is a working, fully grounded verdict on real documents — not breadth of features.

Primary goals

1. Return a contestability verdict on a real rejection letter and real policy wording, with every statement traceable to a source.
2. Never assert a clause or provision that is not present in the documents supplied or in the coded IRDAI rulebook.
3. Produce an appeal document the user could actually file, without a developer present.

Secondary goals

1. Deliver the verdict and the appeal in Hindi as well as English.
2. Correctly return a Weak verdict where the rejection is valid, and explain why.
3. Keep the whole flow usable by someone who has never seen the product, with no account and no onboarding.

Hackathon goals

1. Demonstrate the full flow live on documents not seen before by the judges.
2. Answer the hallucination question with architecture rather than assurance.
3. Show the team split working: two members building, three owning the rulebook, design and testing.

6. Non-Goals

These are excluded from the MVP by decision, not by oversight. The deck states them on the record. **[Deck]**

Not building	Why
User accounts, login, saved case history	Adds auth, storage and privacy surface with no demo value
Any dashboard or analytics view	Nothing to aggregate in a single-session tool
Motor, life or travel insurance	Each needs its own rulebook; health only for the MVP
Mobile application	Responsive web is sufficient to demonstrate the flow
Insurer or TPA-facing portal	Different product for a different party
Automated filing or submission to any portal	The user files their own documents; automating this would be acting on their behalf in a legal process

Not building	Why
Payments or subscription plumbing	Free for individuals in the MVP
Case-precedent search over Ombudsman awards	Awards are released only to the individual complainant, so no public corpus exists to search
More than one regional language	Hindi proves the capability; more languages are configuration, not capability

Not attempted at all, at any phase. Predicting whether the insurer will concede; estimating a settlement amount; representing the user; advising on litigation. Pratkar prepares documents the user files themselves.

7. Product Scope

MVP — must be working on 20 September 2026 [Deck]

- Dual document upload: rejection letter (photo or PDF) and policy wording PDF
- Extraction of insurer, claim value, stated rejection ground, cited clause reference, policy tenure
- Verbatim retrieval of the cited clause from the user's policy, with page reference
- Deterministic checks against coded IRDAI provisions
- Contestability verdict: Strong, Moderate or Weak, with reasons
- Evidence trail: every output statement traceable to a policy page span or a named provision
- One generated appeal document, downloadable
- Hindi output for the verdict and the letter

Phase 2 — after the hackathon

- The remaining appeal documents: grievance-portal complaint text and the Ombudsman complaint form
- Escalation clock with the statutory deadlines surfaced and tracked
- Additional regional languages beyond Hindi
- Saved cases and a returning-user account
- Accuracy benchmarking against a larger labelled case set
- A reviewed-by-a-human tier for the secondary users in §4

Future scope — long term

- Motor and life claim rejections on the same architecture, with their own rulebooks

- A precedent layer, if a lawful source of decided cases becomes available
- Aggregate reporting on which rejection grounds do not survive examination
- An interface for agents, TPA desks and NGOs handling rejections in volume

Scope boundary. Anything a mentor or judge requests that is not in the MVP list becomes a roadmap item, not a task. This rule is in force from now until 20 September. **[Deck]**

8. Functional Requirements

Every P0 below must work on 20 September; the MVP is not demonstrable without all of them.

Priorities: **P0** critical, MVP blocker · **P1** high, MVP if time allows · **P2** medium, Phase 2 · **P3** future.

ID	Feature	Description	User	Priority	Acceptance criteria
FR-01	Dual document upload	Accept a rejection letter as photo or PDF and a policy wording PDF in one session	Policyholder	P0	Both files accepted; a 40–80 page policy PDF uploads without timeout; unsupported type rejected with a plain-language message
FR-02	Rejection letter extraction	Extract insurer name, claim value, stated rejection ground and the cited clause reference	System	P0	On the three demo letters, all five fields extracted correctly; any field not found is returned as null, never guessed
FR-03	Policy parsing and indexing	Parse the policy PDF into page-referenced, retrievable text spans	System	P0	Every retrieved span carries its page number; a scanned policy is reported as unsupported rather than silently mis-parsed
FR-04	Verbatim clause retrieval	Locate the cited clause in the user's own policy and return its exact text	System	P0	Returned text matches the PDF character for character; the page number resolves to the correct page
FR-05	Deterministic rule engine	Evaluate the claim facts against coded IRDAI provisions before any model reasoning	System	P0	Each coded rule returns pass / fail / not-applicable with the provision cited; results reproducible across runs on identical input

ID	Feature	Description	User	Priority	Acceptance criteria
FR-06	Contestability verdict	Return Strong, Moderate or Weak with the reasons behind it	Policyholder	P0	Verdict displayed with at least one cited reason; the three demo cases return the expected verdicts, including one Weak
FR-07	Evidence trail	Link every output statement to a policy page span or a named IRDAI provision	Policyholder	P0	No statement in the verdict lacks a source; clicking a citation shows the underlying text
FR-08	Quote-or-abstain enforcement	Block any generated assertion not grounded in the supplied documents or the rulebook	System	P0	On an ungroundable question the system returns "not determinable from the documents provided"; verified by a deliberate negative test case
FR-09	Appeal letter generation	Produce a grievance-officer letter reflecting the verdict and its evidence	Policyholder	P0	Letter generated for a Strong verdict, contains the claim reference and the cited grounds, and is downloadable
FR-10	Hindi output	Render the verdict and the appeal letter in Hindi	Policyholder	P1	A language toggle switches both; Devanagari renders correctly in the downloaded file
FR-11	Vague or clause-less rejection	Where the letter cites no clause, say so and draft a written request for the insurer's grounds	Policyholder	P1	System does not fabricate a clause; produces a request-for-grounds letter instead
FR-12	Weak verdict path	Where the rejection is valid, say so and explain why, with no appeal generated	Policyholder	P0	The designated Weak demo case returns Weak with cited reasoning and offers no appeal letter
FR-13	Session document disposal	Discard uploaded documents at end of session unless the user saves them	Policyholder	P1	Files removed from storage after the session; behaviour stated to the user before upload
FR-14	Non-advice notice and deadline	Display the not-legal-advice notice and the applicable statutory deadline with every verdict	Policyholder	P1	Notice and deadline visible on the verdict screen and reproduced in the generated letter

ID	Feature	Description	User	Priority	Acceptance criteria
FR-15	Remaining appeal documents	Generate grievance-portal complaint text and the Ombudsman complaint form	Policyholder	P2	Both generated and downloadable, populated from the same extracted facts
FR-16	Escalation clock	Show the escalation ladder and the remaining time at each stage	Policyholder	P2	Ladder rendered with dates computed from the rejection date

Deferred by decision. Case-precedent retrieval is not specified as a requirement at any priority — see §6 and §12.

9. User Flows

The primary flow is four steps and requires no account. **[Deck]**

Flow A — contestable rejection (the main path)

flowchart TD

```

A[Upload rejection letter<br/>+ policy wording] --> B[Extract claim facts]
B --> C[Retrieve cited clause<br/>from the policy]
C --> D[Run IRDAI rule checks]
D --> E[Merge and cite]
E --> F{Verdict}
F -->|Strong or Moderate| G[Show evidence trail]
G --> H[Generate appeal letter]
H --> I[Switch to Hindi<br/>and download]

```

1. User opens the site and uploads both documents. No sign-up.
2. System extracts insurer, claim value, stated ground, cited clause and policy tenure, and shows them for confirmation.
3. System retrieves the cited clause verbatim from the user's policy and runs the deterministic rule checks.
4. System returns a verdict with the evidence trail: the quoted clause, the applicable provision, and the reasoning that links them.
5. User opens the appeal letter, switches language if they want, and downloads it.
6. User files it themselves. The non-advice notice and the deadline are on screen and in the letter.

Flow B — rejection is valid (the Weak path)

Steps 1–4 are identical. At step 4 the verdict is Weak: the system states that the rejection stands, cites the clause and provision that make it valid, and does not generate an appeal. The user is told what would change the position — for example, a document that was not submitted.

Flow C — rejection letter cites no clause

At step 2 no clause reference is found. The system does not infer one. It reports that the insurer has not stated a ground in the required form, and generates a written request for the insurer's grounds instead of an appeal. **[Rec]**

Flow D — documents cannot be processed

A scanned or illegible policy PDF, or a rejection letter photo too poor to read, ends the flow with a specific message naming which document failed and what would fix it. The system never proceeds on a partial parse. See §14.

10. System Workflow

Two reasoning paths run independently and are merged only at the end, so neither can invent an answer the other cannot support. **[Deck]**

```
flowchart LR
    I1[Rejection letter] --> EX[Extraction]
    I2[Policy wording] --> EX
    EX --> A[Path A<br/>Rule engine]
    EX --> B[Path B<br/>Clause retrieval]
    A --> M[Merge and cite]
    B --> M
    M --> O[Verdict, evidence,<br/>appeal, language]
```

Stage	Input	Processing	Output
Ingestion	Two uploaded files	Type and size validation; text extraction from the policy PDF; vision extraction from a photographed letter	Raw text plus page map

Stage	Input	Processing	Output
Extraction	Raw text	Identify insurer, claim value, stated ground, cited clause reference, policy tenure, relevant dates	Structured claim record
Path A — rule engine	Structured claim record	Evaluate coded IRDAI provisions. No model involved	Pass / fail / not-applicable per rule, each citing its provision
Path B — clause retrieval	Structured claim record, indexed policy	Retrieve the cited clause and the clauses it depends on, verbatim with page references	Quoted spans with page numbers
Merge and cite	Both path outputs	Combine into a verdict; reject any assertion lacking a source	Verdict, reasons, evidence trail
Generation	Verdict and evidence	Draft the appeal from the cited grounds; translate on request	Appeal document, English or Hindi

Ordering rule. Path A runs first and its results are binding. Where a coded provision settles the question — for example, a rejection on non-disclosure grounds after the moratorium has run — the verdict follows the rule, and Path B supplies supporting quotation rather than judgement. **[Rec]**

Merge rule. The merge step drops any statement that cannot name a policy page span or a provision. A verdict may therefore be returned with fewer reasons than the model proposed, and that is the intended behaviour.

11. AI/ML Requirements

The model is never allowed to make the legal claim. This is the product's stated innovation and it constrains every requirement below. **[Deck]**

No model is trained. There is no training set, no fine-tuning and no custom model. The system orchestrates existing models under hard constraints. **[Deck]**

Component	Type	Role	Not its role
Rule engine	Deterministic code, no model	Decide questions that published IRDAI provisions settle outright	Anything requiring interpretation
Vision extraction	Vision-capable LLM	Read a photographed rejection letter into text	Interpreting what the letter means
Field extraction	Long-context LLM	Pull the structured claim record out of the letter	Inferring a field that is absent
Clause retrieval	Embeddings + long-context LLM	Locate and quote the relevant clauses from the user's policy	Paraphrasing or summarising a clause
Verdict reasoning	Long-context LLM	Explain how the quoted clause and the provision apply to the facts	Asserting anything it cannot quote
Appeal drafting	Long-context LLM	Compose the letter from grounds already established	Introducing a new ground
Translation	Bhashini API, IndicTrans2 fallback	Render verdict and letter in Hindi	Changing substance in translation

Prompting requirements

1. Every reasoning call receives the retrieved policy spans and the rule-engine results as context. The model is not asked to recall insurance law from memory.
2. The output schema requires a source for each assertion: a page span identifier or a provision reference. An assertion without one is invalid output, not a lower-confidence output.
3. The abstain path is explicit. The instruction to return "not determinable from the documents provided" is part of the required schema, not a fallback in the prose.
4. Extraction and reasoning are separate calls. Extraction returns fields or null; it is never asked to reason about them.

Evaluation

- Ground truth is the labelled case set in §12, run by Dhiraj against every build.
- Metrics: field extraction accuracy, clause retrieval correctness, verdict agreement with the labelled outcome, and grounding violations — the count of output statements lacking a valid source, which must be zero.

- Reporting rule: any accuracy figure quoted publicly carries its sample size. **[Rec]**

Confidence and uncertainty. The verdict is a three-level judgement, not a probability. No numeric confidence score is displayed, because a percentage on a legal judgement implies a calibration the system does not have. Moderate exists precisely to carry genuine uncertainty. **[Rec]**

Failure handling. A model call that fails, times out or returns schema-invalid output is retried once, then falls back to the second provider. If both fail, the system returns the deterministic rule-engine results alone and states plainly that the interpretive analysis is unavailable. It never returns a verdict built on a partial pipeline. See §14.

12. Data Requirements

The user supplies the data the system reasons over. There is no corpus to acquire and no dataset to license.

Data	Source	Status	Owner
Policy wording PDFs	Published by insurers; supplied by the user at runtime	Confirmed — public	Navya
IRDAI regulatory provisions	IRDAI Master Circular on Health Insurance, IRDAI/HLT/CIR/PRO/84/5/2024 dated 29.05.2024	Confirmed — public	Navya
Standardised non-payable items	IRDAI/HLT/REG/CIR/193/07/2020, 146 items across four lists	Requires validation — see open item below	Navya
Real rejection letters	Collected by the team from family, friends, agents and TPA desks	Requires collection — the hard dependency	Navya
Insurance Ombudsman awards	Released only to the individual complainant	Not available — no public corpus; out of MVP scope	—

Open item requiring resolution before the rule engine ships. Confirm whether the 2020 standardisation circular survived the repeal schedule of the 2024 Master Circular. If it did not, the 146-item non-payable list must be sourced from the superseding instrument or that rule set must be withdrawn from the engine. **[TBD]**

Structured claim record — the system's core object

Field	Type	Source	Nullable
<code>insurer_name</code>	string	Rejection letter	No
<code>policy_number</code>	string	Rejection letter	Yes
<code>claim_reference</code>	string	Rejection letter	Yes
<code>claim_amount</code>	decimal	Rejection letter	Yes
<code>rejection_date</code>	date	Rejection letter	No
<code>stated_ground</code>	text	Rejection letter	No
<code>cited_clause_ref</code>	string	Rejection letter	Yes — drives Flow C
<code>policy_inception_date</code>	date	Policy or letter	Yes — required for moratorium check
<code>continuous_months</code>	integer	Derived	Yes

Output object. Verdict level; an ordered list of reasons, each carrying either a page span reference or a provision reference; the generated appeal text; the language rendered.

Preprocessing. Policy PDFs are split into page-scoped chunks with page numbers preserved through the whole pipeline, because the page reference is what makes the evidence trail verifiable. Photographed letters are deskewed before vision extraction.

[Rec]

Storage. Uploaded documents live only for the session and are discarded afterwards unless the user saves them (FR-13). The coded rulebook is version-controlled in the repository, not in a database.

Quality concerns

1. Scanned image-only policy PDFs cannot be page-span cited reliably and are rejected rather than processed.
2. Insurer letters vary widely in format; extraction accuracy will differ by insurer, and the labelled set should cover more than one.
3. The labelled case set will be small. Any accuracy claim must state its sample size.

13. Non-Functional Requirements

Targets below are proposed for the MVP and sized to a demo, not to production load.

[Rec]

Attribute	Requirement	Rationale
Performance	Full analysis returns within 90 seconds for a 60-page policy; upload acknowledged within 2 seconds	The demo must not stall in front of judges; a progress state is shown throughout
Demo performance	The three designated demo cases return from cache in under 5 seconds	Removes live model latency from the critical demo path
Scalability	Single concurrent user for the MVP; no horizontal scaling required	A hackathon demo is one session at a time. Stated openly rather than overclaimed
Reliability	A failed model call retries once, then fails over to the second provider	No single provider is a single point of failure on demo day
Availability	Deployed and reachable from 18 September, not first deployed on the day	Deploy early so integration problems surface before they matter
Security	No credentials in client code; secrets in environment variables only	See §15
Privacy	Documents discarded at end of session; no training use; no third-party sharing beyond the model provider call	Health and financial documents; see §15
Maintainability	The IRDAI rulebook is data, not code branches, so a non-developer can read and extend it	Navya owns the rulebook and is not writing application code
Usability	The full flow is completable by someone who has never seen the product, with no instructions	Tested by having a non-team member run it before 19 September
Accessibility	Devanagari renders correctly on screen and in generated documents	Hindi output is a stated MVP feature

Attribute	Requirement	Rationale
Observability	Every analysis logs its stage timings, the rules fired, and any grounding violation	Needed to answer "how do you know it worked" with evidence

Explicitly not required for the MVP. Uptime guarantees, load testing, multi-region deployment, disaster recovery, rate limiting beyond provider defaults. These belong to a production version, not to a 72-hour build.

14. Error Handling

The governing rule: when the system cannot complete an analysis correctly, it says so. It never returns a degraded verdict that looks like a full one.

Condition	Behaviour	User sees
Unsupported file type	Reject at upload, before any processing	"We can read PDF, JPG and PNG. This file is a .docx — please upload the PDF version."
Policy PDF is scanned images only	Stop; do not attempt page-span citation	"We cannot cite pages from a scanned policy. Please upload the digital PDF your insurer issued."
Rejection letter photo unreadable	Stop at extraction; name the problem	"We could not read this image. Try better lighting, or upload the PDF."
Cited clause reference absent	Proceed on Flow C; do not infer a clause	"Your insurer has not stated which clause it relied on. Here is a letter requesting those grounds."
Cited clause not found in the policy	Report the mismatch — itself a possible ground	"The clause your insurer cited does not appear in this policy. That is worth raising."
Required field missing for a rule	Skip that rule; state which check could not run	"We could not check the moratorium because the policy start date was not found."

Condition	Behaviour	User sees
Model returns schema-invalid output	Retry once, then fail over to the second provider	Progress indicator continues; no error unless both fail
Both model providers fail	Return deterministic rule results only	"Regulatory checks completed. The detailed clause analysis is unavailable right now."
Grounding violation detected at merge	Drop the offending statement; keep the rest	Fewer reasons shown; no visible error
No statement survives grounding	Return no verdict	"We could not reach a conclusion we can evidence from your documents."
Translation service unavailable	Fall back to the offline translation path; if that fails, serve English	"Hindi is unavailable right now — showing English."
Database unavailable	Continue the analysis in memory for the session	No visible error; saving is disabled
Upload exceeds size limit	Reject with the actual limit named	"This file is 48 MB. The limit is 25 MB."

Rule. No error message says "something went wrong." Every message names what failed and what the user can do about it. **[Rec]**

15. Security Requirements

The product handles health and financial documents belonging to people in difficulty. That sets the bar, even for a hackathon MVP.

ID	Requirement	Priority
SEC-01	All traffic over HTTPS; no document transmitted over plain HTTP	P0
SEC-02	API keys and provider credentials in environment variables only; never in client code or the repository	P0
SEC-03	Uploaded documents discarded at end of session unless the user explicitly saves them	P0

ID	Requirement	Priority
SEC-04	Uploads validated by type, size and content before processing	P0
SEC-05	No document content used for model training; provider settings configured accordingly	P0
SEC-06	Documents never shared with any third party beyond the model provider call required to process them	P0
SEC-07	Consent shown before upload, stating what is processed, by whom and for how long	P1
SEC-08	Logs record stage timings and rule outcomes, never document content or extracted personal data	P1
SEC-09	Generated documents served to the requesting session only, no public URLs	P1
SEC-10	Prompt-injection defence: policy and letter text treated as data, never as instruction	P1

On SEC-10. A policy wording or rejection letter is a document written by another party. If one contained text resembling an instruction, the system must not follow it. Retrieved spans are passed as quoted evidence, and the output schema constrains what can be produced from them. **[Rec]**

Not in MVP scope. Penetration testing, formal threat modelling, a certified compliance posture, or a data-processing agreement. The MVP has no user accounts, so there are no credentials to protect and no session hijacking surface beyond the single active session. A production version would need all of these. **[Rec]**

16. Success Metrics

All numeric targets below are **proposed** by this document, not established by the deck or by any measurement taken so far. They exist to make the MVP falsifiable. **[Rec]**

Metric	Definition	Proposed MVP target	Measured by
Grounding violations	Output statements lacking a valid page span or provision reference	Zero. This is the one non-negotiable target	Automated check on every analysis

Metric	Definition	Proposed MVP target	Measured by
Field extraction accuracy	Correctly extracted fields ÷ total fields, across the labelled set	Report actual with sample size; no target claimed	Dhiraj, against ground truth
Clause retrieval correctness	Cited clause correctly located and quoted verbatim	Report actual with sample size	Dhiraj, manual verification
Verdict agreement	Verdicts matching the labelled outcome	Report actual with sample size	Dhiraj, against ground truth
Abstention correctness	Cases where the system correctly declines rather than guessing	All deliberate negative tests abstain	Dedicated negative test cases
End-to-end latency	Upload to verdict, 60-page policy	Under 90 seconds	Instrumented timing
Unassisted completion	A non-team member completes the flow without help	At least one successful run before 19 September	Observed test

Reporting discipline. No accuracy percentage is stated in a deck, a demo or a conversation with judges without its sample size attached. A figure from a set of twelve cases is presented as "11 of 12," never as "92%." **[Rec]**

Post-hackathon metrics, not measurable now. Appeals actually filed; appeals succeeding; amount recovered; adoption by the secondary users in §4. These require users and time, and no target is proposed for them here. **[TBD]**

17. MVP Acceptance Criteria

The MVP is accepted when all eleven statements below are true, verified by a run-through on 19 September with no developer intervention.

- ☐ A real rejection letter and a real policy wording PDF upload successfully in one session
- ☐ The structured claim record is extracted and displayed for confirmation
- ☐ The cited clause is retrieved verbatim from the policy and shows its page number
- ☐ The deterministic rule engine runs and reports which provisions fired
- ☐ A verdict of Strong, Moderate or Weak is returned with at least one cited reason
- ☐ Every statement in the verdict resolves to a page span or a named provision

- ☐ A deliberate ungroundable case returns "not determinable" rather than an answer
- ☐ A Strong verdict produces a downloadable appeal letter containing the claim reference and cited grounds
- ☐ The designated Weak case returns Weak, explains why, and offers no appeal letter
- ☐ The verdict and letter render correctly in Hindi
- ☐ The non-advice notice and the applicable deadline appear on the verdict screen

Demo readiness, additional to the above

- ☐ Three demo cases pre-cached and returning in under 5 seconds
- ☐ A fourth unseen letter held in reserve for a judge who asks whether it is hardcoded
- ☐ A recorded run of the full flow stored on a local drive
- ☐ The full demo completed twice without a developer touching anything

Not required for acceptance. Every P2 item, the remaining appeal documents, the escalation clock, and any accuracy threshold. A working, honest, fully grounded flow is the bar.

18. Risks and Mitigation

Risk	Probability	Impact	Mitigation
Too few real rejection letters collected — the hard dependency	Medium	Critical	Collect from family, friends, agents and TPA desks. Realistic constructed letters clearly labelled as such are an acceptable fallback for testing, never for accuracy claims. Owner: Navya
Model asserts a clause not in the policy during a live demo	Low	Critical	Quote-or-abstain enforced at generation, not checked afterwards. Demo cases pre-cached. Never run an uncached case on stage. Owner: Sid
The 2020 non-payable list turns out to be repealed	Medium	Medium	Resolve the open item in §12 before coding those rules. If unresolved, ship without that rule set rather than on an uncertain basis. Owner: Navya

Risk	Probability	Impact	Mitigation
Scope creep from mentoring feedback today	High	High	Feature freeze. Any request becomes a roadmap slide. Fix only objections raised twice or more. Owner: Sid
Model provider outage or rate limit on demo day	Medium	High	Second provider configured from the start; demo path served from cache; offline translation fallback. Owner: Pratham
Insurer letters too varied for reliable extraction	Medium	Medium	Cover more than one insurer in the labelled set; report accuracy with sample size; Flow C handles the degraded case. Owner: Dhiraj
Two developers become the bottleneck	Medium	Medium	The three non-code workstreams are file-based and never wait on a merge. Daily standup catches blockage within 24 hours. Owner: all
Judged as "an LLM wrapper"	Medium	High	Lead with the deterministic rule engine and show the two-path architecture before any model output. Owner: Dhiraj on the deck
Legal exposure from generating filings	Low	Medium	Position as document preparation, not representation. Non-advice notice and the statutory deadline on screen and in the letter. User files it themselves. Owner: Sid
Hindi rendering breaks in the generated document	Medium	Low	Test Devanagari in the download path specifically, not just on screen. Owner: Parnika
Demo looks plain next to a rival dashboard	Medium	Medium	Invest design time in the evidence-trail screen, which is both the trust mechanism and the visual. Owner: Parnika

19. Future Roadmap

Stage	Scope	Gate to the next stage
Hackathon MVP — by 20 Sep 2026	Health claims. Verdict, evidence trail, one appeal letter, Hindi. Single session, no accounts	The MVP acceptance criteria in §17 are met
Phase 2 — post-hackathon	Full appeal pack, escalation clock, saved cases, more languages, accuracy benchmarked on a larger labelled set	A measured accuracy figure the team is willing to publish with its sample size
Production version	Accounts, persistence, consent records, DPDP-aligned data handling, monitoring, a reviewed tier for agents and TPA desks	Real users filing real appeals, and evidence that they succeed
Scale version	Motor and life rulebooks; interfaces for agents, TPA desks and NGOs handling volume; aggregate reporting on rejection grounds that do not survive examination	Commercial demand from the secondary users in §4

Sequencing principle. Breadth of insurance lines comes last, not first. Each new line needs its own regulatory rulebook, and the rulebook is what makes the product defensible. Adding motor claims before health is proven would multiply the surface without strengthening the core. **[Rec]**

Precedent, if it ever becomes possible. A precedent layer would materially improve verdict quality, but Ombudsman awards are released only to the individual complainant, so there is no lawful public corpus to build on today. This stays out of the roadmap until that changes. **[Deck]**